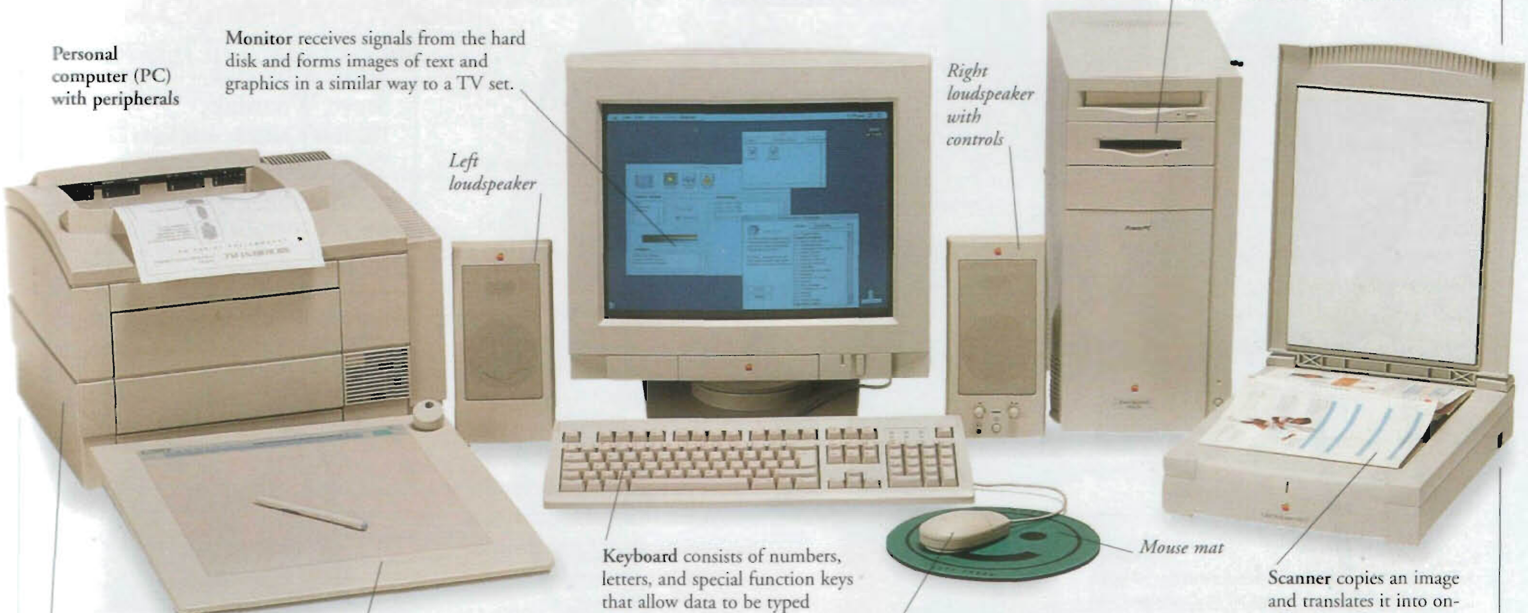
COMPUTERS

WITH LIGHTNING SPEED, a computer carries out millions of calculations each second. Sets of instructions called programs tell the computer what to do. The hard-disk unit is the heart of a computer. It contains the central processing unit (CPU), which controls all of the operations of the computer. The hard-disk unit, monitor, keyboard, and other connected devices are called hardware. The programs that enable it to function and carry out specific tasks are known as software.

Personal computer

A personal computer (PC) is a compact computer that can be operated by only one person at a time. It consists of a hard-disk unit, to which hardware called peripherals is connected so that data can be either input or output. Keyboards and printers are examples of peripherals. Interfaces are electronic circuits that allow the hard-disk unit to communicate with each peripheral.



Personal computer (PC) with peripherals

Monitor receives signals from the hard disk and forms images of text and graphics in a similar way to a TV set.

Left loudspeaker

Right loudspeaker with controls

Hard-disk unit contains the memory, the CPU, and the disk drives.

Keyboard consists of numbers, letters, and special function keys that allow data to be typed directly into the computer.

Mouse mat

Scanner copies an image and translates it into on-off pulses of electricity that are fed into the computer, so that the image can be displayed on the monitor screen.

Printer receives data from hard disk and produces print-out of documents and graphics.

Graphics tablet enables images to be "drawn" on the monitor screen, as a pen-like device moves over its surface.

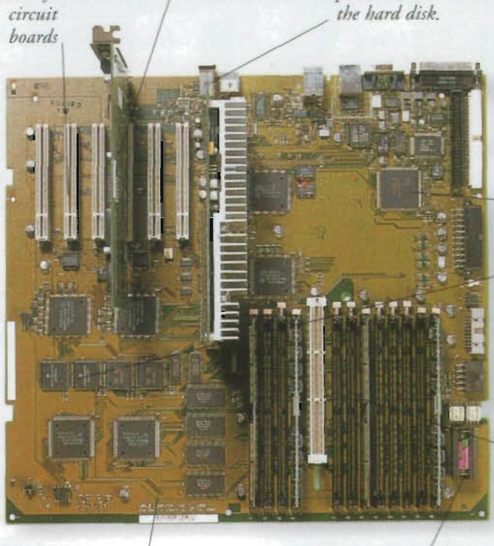
Mouse controls pointer on screen; inside the mouse is a ball that rotates as the mouse moves, and the ball's movement sends signals to the computer.

Motherboard

A motherboard is a large circuit board in the hard-disk unit to which the computer's key electronic components are attached. These components are linked together by strips of metal called "buses" on the underside of the motherboard. Also attached to the motherboard are the interfaces that link the hard-disk unit to the peripherals, as well as expansion slots, to which other circuit boards can be added to improve the computer's performance or capabilities.

Memory

A computer's electronic memory allows it to "remember" how to function. There are two parts to the memory: the random access memory (RAM) and the read-only memory (ROM). Both consist of circuits called microprocessors, or silicon chips.



PC motherboard

Expansion slots for extra circuit boards

Video card controls operation of monitor screen.

Sockets called ports allow peripherals such as a modem or printer to connect to the hard disk.

Central processing unit (CPU)

ROM (read-only memory) chips store important programs, such as the disk-operating system, whose content cannot be changed.

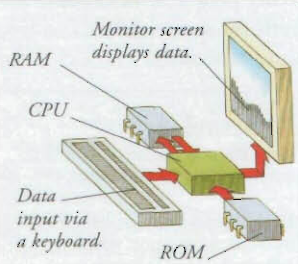
RAM (random-access memory) chips store data fed into the computer on disks or via the keyboard, which can then be retrieved and changed as desired.

Buses carry signals around the computer.

Battery controls computer's internal clock.

Charles Babbage

English mathematician Charles Babbage (1791–1871) built a mechanical computer called the Difference Engine that consisted of hundreds of gear wheels. It could do complicated sums more quickly than doing the same calculations by hand.



Central processing unit (CPU)

The CPU is a single microprocessor that holds a large number of circuits. The CPU receives data from the ROM, RAM, and keyboard. It sends data to the RAM for storage, and to output devices, such as the monitor.